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Master Thesis in Big Data Management and Analytics

Trustworthy Data-Aware Process Simulation for Decision Support in Container Terminal Truck Operations

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Abstract

The exponential scaling of artificial neural networks (ANNs) [?] has driven remarkable performance breakthroughs [1? ?]. Yet this success is increasingly bottlenecked by the unsustainable energy demands and memory overhead of these computation-intensive architectures [? ? ?]. *Spiking Neural Networks* (SNNs) present an energy-efficient alternative to traditional ANNs by mimicking the brain's biological approach to information processing [? ?]. However, this energy efficiency, achieved through the temporal binarization of information, relies on step functions that possess zero derivatives almost everywhere. Consequently, standard gradient-based training methods become intractable. Current state-of-the-art (SOTA) methods bypass this issue by utilizing surrogate gradients (SG-BPTT) [?], which rely on smooth approximations of the spikes during backpropagation through time (BPTT), therefore introducing biases in the gradient estimates. To overcome these inherent limitations, this thesis presents a gradient-free optimization framework for SNNs based on the *Alternating Direction Method of Multipliers* (ADMM) [? ? ?].

This work presents a modular, open-source Python library that establishes ADMM as a gradient-free alternative to the standard backpropagation-through-time (BPTT) algorithms which conventionally relied on automatic differentiation frameworks like PyTorch and TensorFlow. Engineered to facilitate experimentation and serve as a foundation for future research, the framework natively supports both non-spiking ANNs and SNNs. It extends the algorithmic formulation of ADMM to support convolutional layers, various loss functions, and modern training heuristics such as flexible iteration schedules and toggleable Lagrange multipliers. Furthermore, to ensure computational efficiency, the library leverages established numerical techniques, including Fast Fourier Transforms (FFTs) [?] and the Woodbury matrix identity [?], alongside multibatch processing.

Finally, we empirically validate the proposed framework through extensive experimentation. We demonstrate significant improvements over the SOTA ADMM in both training accuracy and computational efficiency, a finding that is then discussed through an extensive ablation study. Furthermore, we explore and compare various architectures in relation to gradient-based methods, providing empirical support for this line of research. We conclude the experimentation by introducing a decentralized parallelization strategy, providing a scalable alternative to the standard master-worker consensus ADMM architecture, and demonstrating its distributive capabilities across different setups.

Contents

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Listing of acronyms

ANN .... Artificial Neural Network
CNN .... Convolutional Neural Network
RNN .... Recurrent Neural Network
SNN .... Spiking Neural Network
CSNN .... Convolutional Spiking Neural Network
LIF .... Leaky Integrate-and-Fire
ADMM .... Alternating Direction Method of Multipliers
SGD .... Stochastic Gradient Descent
BPTT .... Backpropagation Through Time
SG-BPTT ...... Surrogate Gradient Backpropagation Through Time
STDP .... Spike-Timing-Dependent Plasticity
SOTA .... State-of-the-Art
FFTs .... Fast Fourier Transforms
KCL .... Kirchhoff's Current Law
L-BFGS .... Limited-memory Broyden–Fletcher–Goldfarb–Shanno
SVHN .... Street View House Numbers
MNIST .... Modified National Institute of Standards and Technology
N-MNIST ..... Neuromorphic-MNIST
CPU .... Central Processing Unit
GPU .... Graphics Processing Unit
HPC .... High-performance computing
NCCL .... NVIDIA Collective Communications Library

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Introduction

Container terminals are critical nodes within global supply chains and play a key role in ensuring the efficient movement of goods between maritime and hinterland transportation networks. Increasing container volumes, larger vessel sizes, and growing operational complexity require terminal operators to make numerous tactical and operational decisions under uncertain conditions. Examples include truck dispatching, yard block allocation, resource utilization, congestion management, and the handling of disruptions such as equipment failures or temporary infrastructure outages. Traditional experimentation in real terminal environments is often costly, disruptive, or operationally infeasible. Therefore, simulation has become a widely used decision-support technique in terminal planning and operations research. [PROSIT (2026) | PDF], [github.com], [scholar.google.com]

1.1 Motivation

Simulation is a widely used decision-support technique in container terminal planning and operations research. Due to the operational continuity of container terminals, experimentation in real environments is generally costly, disruptive, and associated with considerable operational risk. Consequently, simulation models are frequently employed to evaluate infrastructure investments, operational policies, scheduling approaches, truck appointment systems, and terminal redesign alternatives before implementation. Traditional approaches predominantly rely on manually developed Discrete Event Simulation (DES) models, agent-based models, or digital twins that require substantial modeling effort, expert knowledge, and continuous calibration to accurately represent real-world terminal operations [2, 3, 4?]. In parallel, the Process Mining community has investigated methods for automatically extracting process knowledge from event logs generated by operational information systems. One long-standing research objective has been the automated construction of simulation models directly from process execution data. Early work demonstrated that simulation-relevant information, including control-flow behavior, activity durations, arrival patterns, resource allocations, and routing probabilities, can be automatically discovered and incorporated into executable simulation models [5]. Recent advances have extended this idea toward *data-aware process simulation*. Instead of relying solely on control-flow information and static branching probabilities, these approaches incorporate contextual process attributes, case-specific information, process states, resource characteristics, and decision logic into automatically discovered simulation models [6, 7, 8, 9]. Furthermore, recent frameworks such as ProSiT advocate explainable and configurable simulation models based on probabilistic white-box predictors, with the explicit goal of supporting what-if analyses while maintaining model transparency and interpretability [10]. At the same time, research has increasingly highlighted the limitations of black-box simulation approaches. While machine learning and deep learning methods are able to learn complex behavioral patterns, they often provide limited transparency and are difficult to inspect, modify, and validate when used for operational decision support. Consequently, recent studies advocate hybrid and white-box approaches that preserve explainability while exploiting data-driven behavioral discovery [11, 10]. Despite these advances, empirical evidence regarding the applicability of automatically discovered simulation models in logistics systems remains limited. Existing evaluations primarily focus on business-process environments such as healthcare, financial services, and administrative processes [12, 8]. In contrast, container terminal truck operations are characterized by physical resource constraints, congestion effects, infrastructure dependencies, spatial interactions, and highly dynamic process contexts. Whether automatically discovered data-aware simulation models can adequately represent these characteristics and provide trustworthy decision support remains largely unexplored. This thesis addresses this research gap by investigating the applicability, realism, robustness, and decision-support capability of data-aware process simulation models discovered from real container terminal truck operation event logs.

1.2 Problem Statement

Container terminal operators continuously face decisions regarding capacity management, truck handling processes, yard utilization, operational policies, and infrastructure changes. Simulation models are frequently employed to evaluate such decisions because experimentation in the real terminal environment is often infeasible or associated with significant operational risks. However, most existing container terminal simulation studies rely on manually developed simulation models. These models require substantial domain expertise and modeling effort and are often costly to maintain as terminal operations evolve. Furthermore, manually constructed models may introduce subjective assumptions regarding process behavior, resource interactions, and routing logic [3, 4]. Data-aware process simulation offers a fundamentally different paradigm. Instead of manually designing simulation behavior, the simulation model is automatically discovered from historical event logs. This approach promises improved scalability, reduced modeling effort, and closer alignment with observed process behavior [13, 9, 8]. However, the practicality of this approach implicitly depends on several assumptions. It must be demonstrated that automatically discovered models can adequately reproduce observed operational behavior, that they remain interpretable and configurable, and that they provide sufficiently robust simulation outcomes to support decision-making. While previous studies have demonstrated the technical feasibility of data-aware simulation in business-process environments [8, 12], it remains unclear whether comparable results can be achieved in logistics systems characterized by congestion effects, resource competition, and highly context-dependent behavior. Therefore, a systematic evaluation of the trustworthiness and limitations of automatically discovered data-aware simulation models in container terminal truck operations is required.

1.3 Research Objectives

The primary objective of this thesis is to evaluate whether automatically discovered data-aware process simulation models can provide trustworthy and robust decision support in the context of container terminal truck operations. More specifically, the thesis aims to:

- assess the ability of automatically discovered simulation models to reproduce observed operational behavior;

- investigate the modeling challenges that emerge when applying data-aware simulation discovery techniques to container terminal processes;

- analyze how alternative modeling assumptions affect simulation outcomes and resulting decision recommendations;

- evaluate the suitability of discovered simulation models for operational what-if analyses and previously unobserved scenarios.

The study focuses on truck handling operations within a container terminal environment and utilizes real operational event data as the basis for process discovery and simulation model generation.

1.3.1 Research Contributions

The expected scientific contributions of this thesis are fourfold.

1. **Domain Evaluation Contribution**
   The thesis provides one of the first evaluations of data-aware process simulation in the domain of container terminal truck operations.

2. **Methodological Contribution**
   The thesis identifies domain-specific challenges and limitations that emerge when transferring data-aware simulation approaches from traditional business-process environments to logistics systems with physical constraints and congestion effects.

3. **Robustness and Trustworthiness Contribution**
   The thesis investigates how alternative plausible specifications of resources, arrival processes, and contextual attributes influence simulation outcomes and decision-support recommendations.

4. **Decision-Support Contribution**
   The thesis evaluates the suitability of automatically discovered simulation models for operational what-if analyses involving scenarios not present in the original event log.

1.4 Research Questions

To achieve these objectives, the thesis addresses the following research questions:

1. To what extent can a data-aware process simulation model discovered from historical container terminal event logs reproduce observed operational behavior?

2. Which modeling challenges and limitations emerge when applying data-aware simulation discovery techniques to container terminal truck handling operations?

3. To what extent do alternative plausible specifications of discovered resources, arrival processes, and contextual process attributes affect simulation outcomes and decision recommendations?

4. To what extent can automatically discovered simulation models provide reliable decision-support recommendations for operational scenarios not represented in the historical event log? endenumerate

1.5 Thesis Outline

The remainder of this thesis is structured to guide the reader from the theoretical foundations of process simulation and process mining to the design of the research methodology, the implementation of the simulation models, and the empirical evaluation of their applicability to container terminal operations.

Following this introduction, Chapter 2 establishes the theoretical foundation of the thesis. It introduces the operational characteristics of container terminals and truck handling processes, reviews the fundamentals of Process Mining, and discusses the evolution of Business Process Simulation from traditional manually developed simulation models towards simulation discovery and data-aware process simulation approaches. Furthermore, the chapter reviews the current state-of-the-art in explainable and white-box simulation methods and discusses existing approaches for simulation validation and trustworthiness assessment.

Chapter **??** presents a structured review of the related literature. It examines simulation applications in container terminal operations, automated simulation model discovery, data-aware simulation techniques, and recent advances in explainable simulation frameworks. The chapter also discusses existing evaluation approaches and identifies the research gap addressed by this thesis.

Chapter ?? introduces the research methodology adopted in this work. It describes the case study environment, the underlying event data, the preprocessing pipeline, and the procedures used to construct process mining artifacts and simulation-ready datasets. Furthermore, the chapter presents the evaluation framework used to assess the quality, robustness, and trustworthiness of the generated simulation models.

Chapter ?? presents the discovery and configuration of the data-aware simulation models used throughout the study. It describes how control-flow structures, resource behavior, process-state information, arrival processes, and contextual attributes are extracted from the event logs and incorporated into executable simulation models. Additionally, the chapter discusses the explainability characteristics of the employed white-box simulation approach.

Chapter ?? details the experimental setup and empirical evaluation. It introduces the validation methodology, the performance indicators used to compare real and simulated behavior, and the experimental scenarios designed to assess simulation quality. Particular attention is given to the evaluation of behavioral realism, model robustness under alternative assumptions, and the quality of decision-support recommendations.

Chapter 7 discusses the results obtained throughout the evaluation. It analyzes the strengths and limitations of automatically discovered data-aware simulation models in the context of container terminal truck operations, reflects on the implications for operational decision support, and evaluates the findings in relation to the research questions.

Finally, Chapter 8 concludes the thesis by summarizing its main findings and contributions, discussing methodological and practical limitations, and outlining directions for future research on data-aware process simulation and decision support in logistics systems.

2

Background and Related Work (20)

2.1 Container Terminal Operations

Container terminals represent critical nodes within global supply chains and facilitate the transition between maritime and inland transportation networks. They serve as interfaces between deep-sea shipping and hinterland transportation systems including road, rail, and inland waterway transport. Modern container vessels can transport more than 20,000 containers in a single voyage, requiring container terminals to process large cargo volumes efficiently while maintaining short turnaround times and high service quality. Since their introduction in the 1960s, standardized containers have become the dominant unit for global freight transportation. To ensure comparability across different container sizes, container throughput is commonly measured in Twenty-foot Equivalent Units (TEU). A standard 20-foot container corresponds to one TEU, whereas a 40-foot container corresponds to two TEUs [14]. Container terminals operate under considerable operational pressure. Increasing international trade volumes, larger vessels, and growing demands for rapid cargo handling require terminal operators to coordinate numerous interdependent processes and resources simultaneously. As a result, operational decisions directly affect vessel berth times, truck turnaround times, a terminals throughput, and overall service quality [3, 4]. The empirical context of this thesis is the HHLA Container Terminal Burchardkai (CTB) located in the Port of Hamburg. The

CTB is the largest container terminal in Hamburg and. The terminal handles large volumes of import, export, and transshipment containers while coordinating interactions between vessels, trucks, rail services, storage yards, and container handling equipment. [ACTUAL NUMBERS]

2.1.1 Container Terminals as Complex Logistics Systems

Container terminals, in the research world, serve as highly complex logistics systems due to the large number of interacting resources, transportation modes, and operational dependencies involved in the everyday operations. [3, 4].

Container Flows

Within a seaport container terminal, containers move between multiple transportation modes and storage locations. A container may arrive by deep-sea vessel, feeder vessel, truck, or rail and subsequently leave the terminal using any of these transport modes. Consequently, a terminal continuously coordinates container flows between the seaside and the landside of the supply chain [14]. Hartmann distinguishes four primary transportation modes within container terminal operations: deep-sea vessels, feeder vessels, trucks, and rail services. Together, these modes create numerous possible transport paths through the terminal and generate highly dynamic container flows that must be coordinated efficiently [14]. Most containers entering a terminal undergo at least one temporary storage phase within the container yard before being transferred to their outbound transportation mode. Therefore, terminal operations must coordinate not only transportation processes but also inventory management and storage allocation decisions [3]. Figure: Container Flow through a Container Terminal

Yard Components and Resources

Container terminals consist of multiple operational subsystems that interact continuously. Bett et al. distinguish three primary terminal areas: the quayside, the yard area, and the landside. The quayside includes berths and quay cranes used for vessel loading and unloading. The yard acts as an intermediate storage area for containers, while the landside connects the terminal to trucks and rail operations[15]. For this Thesis relevant is the terminals yard area, that splits into three seperate areas in the case of the CTB in Hamburg:

(a) **Automatic storage Blocks:** The majority of containers is stored in the automatic storage blocks, operated by "Rail Mounted Gantry Cranes" - therefore often called RMG-Blocks. The CTB operates 22 RMG-Blocks with a total capacity of 2231 operative TEU slots and 3 RMGs per Block, operating between the waterside and the landside.

(b) **Manual Yard:** In addition to the automated storage area, containers can be stored in the manually operated yard. Due to its central location within the terminal, this area is frequently used for operationally flexible storage depending on the assigned vessel berth, yard workload, and equipment availability. Container movements are typically performed by manually operated handling equipment.

(c) **Empty Container Yard:** Empty containers are primarily stored in a dedicated empty container yard. Unlike full containers, empty containers are generally not assigned to a specific customer order but are managed as interchangeable inventory within a shipping line's container fleet. Storing empty containers separately prevents them from occupying or being stacked on top of storage positions of full containers that are linked to specific transport orders. The empty container yard is operated using reach stackers, allowing storage heights of up to six containers, whereas loaded containers are generally stacked only up to three containers high.

Trucks can be ordered to pick and deliver their cargo at all of these yard sections, waiting in dedicated parking and waiting positions to be operated by the respective resources. Resource shortages or inefficiencies within one subsystem often show effects throughout the whole terminal and negatively impact other operations [3].

Interdependencies Between Terminal Processes

An important characteristic of container terminals is the strong interdependency of their operational processes. Individual decisions rarely affect only one subsystem. Instead, local disturbances often propagate throughout the entire terminal. For example, delayed vessel arrivals may postpone container unloading operations, which subsequently affects yard occupancy levels. Increased yard congestion can reduce crane productivity and generate longer waiting times for trucks arriving at the terminal. These delays may ultimately result in congestion at terminal gates and reduced service quality for trucking companies [3, 15]. The complexity of these interactions is one of the primary reasons why analytical optimization approaches alone are often insufficient for evaluating operational decisions in container terminals.

2.1.2 Truck Handling Processes

Importance of Truck Operations

Road transportation is a major component of hinterland connectivity in modern container terminals. Although rail and inland waterway transport contribute significantly to container transportation in some regions, trucks remain the dominant transportation mode for many import and export flows, especially As a consequence, terminals must coordinate large volumes of truck arrivals while maintaining acceptable service levels and avoiding excessive congestion [3, 16]. Fluctuating truck arrival patterns can exceed the available processing capacity of gates, yard equipment, and storage areas, resulting in longer waiting times and reduced terminal efficiency [15, 16].

A Typical Truck Visit Process

To regulate truck arrivals and reduce congestion, many container terminals employ a Truck Appointment System (TAS). Within a TAS, trucking companies reserve a predefined time slot before visiting the terminal and specify the container transactions that will be executed during the visit [15, 16]. A typical truck visit begins with the reservation of an appointment slot and the submission of the corresponding transport order. When arriving at the terminal, the truck passes through an automated gate area where Optical Character Recognition (OCR) systems identify containers, vehicle registration plates, and booking information. Depending on the current workload and gate queue lengths, the truck may have to wait before being processed at the physical gate and officially entering the terminal area [15]. After entering the terminal, the truck drives to the assigned container yard location. Depending on the transport order, the truck can perform one or multiple container transactions. These transactions may include:

- delivering an export container,
- receiving an import container,
- delivering or receiving multiple containers while at the same yard location,

A particularly efficient operational pattern is referred to as a *multi move*. In a multi move, a truck delivers one or more export containers and subsequently picks up one or more import containers during the same terminal visit. Multi moves reduce empty truck trips, improve truck utilization, lower emissions, and decrease congestion both inside and outside the terminal, but require efficient routing by the terminal [16]. In some operations, trucks may additionally handle two 20-foot containers during a single operation. Such moves are organized as twin-container transactions, allowing yard equipment to process two containers within a single handling operation and in that way increase operational efficiency while also raising the operational complexity for the terminal. Following the completion of all assigned transactions, the truck proceeds to the exit gate, completes the required departure procedures, and leaves the terminal. The total time spent between entering and leaving the terminal is commonly referred to as the *Truck Turnaround Time (TTT)*, which represents one of the most important performance indicators in truck and overall terminal operations [15].

Terminal Information Systems

Modern container terminals rely on integrated information systems to coordinate operational activities. At the center of these systems is the Terminal Operating System (TOS), which orchestrates the movement and storage of containers across terminal subsystems and manages operational orders throughout the terminal lifecycle. The TOS is typically connected to specialized subsystems including a Gate Operating Systems (GOS), Yard Management Systems, Equipment Control Systems, and appointment management platforms. Together, these systems continuously generate and amanage operational event data whenever trucks arrive, containers are handled, orders are updated, or terminal resources execute work instructions. Each subsystem stores a different subset of operational information. While gate systems primarily record arrival and departure events and details, yard management systems capture container movements, storage locations and equipment execution orders. Consequently, the complete reconstruction of truck visits often requires the integration of multiple operational data sources. For the purposes of this thesis, the Order and Yard Management System serves as the primary data source. Additional information from other operational systems is subsequently used to enrich and validate the resulting event log. The employed data integration process is described in detail in Chapter 3.

Sources of Operational Delays

Truck turnaround times are influenced by various operational factors that function individually or interdependent throughout the terminal processes. Common reasons for delay include:

- congestion at entry and exit gates, also caused and affected by road traffic outside the terminal
- pick up containers that are buried due to unexpected long dwell time on the terminal
- truck traffic congestion within container yard sections and main routes,
- limited availability of yard handling equipment,
- high occupancy levels within storage blocks,
- resource and priority conflicts between truck and vessel operations,
- unexpected equipment failures or maintenance activities

Because these factors are highly interdependent, local disruptions often affect the entire terminal operations. For example, insufficient yard crane availability and a prioritised vessel departure in time can increase truck service time, which affects the trucks waiting times in the yard, which subsequently results in traffic congestion on the terminal area, a lower gate throughput and overall slowed Hinterland performance [3, 15].

Operational Performance Indicators

The performance of truck operations is commonly evaluated using a set of operational key performance indicators (KPIs).

- **Truck Turnaround Time (TTT):** Total time a truck spends inside the terminal between entry and exit.
- **Waiting Time:** Time spent waiting for gate processing, yard equipment, or storage area access.
- **Yard Utilization:** Degree to which storage areas are occupied and operational resources are utilized.

Among these indicators, Truck Turnaround Time is frequently considered the most important measure of truck service quality because it directly reflects the operational efficiency experienced by trucking companies and cargo owners and is therefore one out of four main KPIS for the HHLA Container Terminals.[15, 16].

2.1.3 Simulation-Based Decision Support in Container Terminals

This is now trying to exxplain Why is there a need for better decision support in container terminals in general, why is simulation a common approach for that purpose, and why are traditional simulation approaches insufficient?

The Management of Container Terminals continuously faces operational decision making challenges. Decisions regarding truck appointment policies, resource allocation, equipment scheduling, maintenance planning, and yard management need to be made while operations remain active. Because container terminal processes are very interconnected, local decisions can quickly affect other parts of the system and influence waiting times, resource utilization, throughput, and eventually customer service quality. Although terminal managers often possess extensive operational experience, many decisions involve considerable uncertainty. Changes to operational procedures or capacity allocations may have unintended consequences, create congestion, increase handling times, or reduce service reliability. As a result, decision-making in container terminals increasingly relies on analytical and digital decision-support approaches that allow operational alternatives to be assessed before implementation. SSimulation as a decision support tool in container terminal management has become an interesting and widely adopted research field. The application is reality is still lacking though, due to high complexity, costs, and a missing technical expertise or priority in the terminal teams. Running a Simulation enables managers and researchers to evaluate operational scenarios in a virtual environment without affecting live terminal operations. By reproducing the interactions between trucks, handling equipment, storage areas, and gate processes, simulation can estimate the effects of operational changes before they are implemented in practice. Typical applications of simulation in container terminals include the evaluation of Truck Appointment Systems (TAS), equipment allocation strategies, resource planning, yard management and storage placement strategies. Simulation is especially valuable when analysing congestion effects, resource bottlenecks, and operational trade-offs that are difficult to evaluate based on purely critical analytical thinking. A variety of studies have demonstrated the usefulness of simulation for improving truck turnaround times, resource utilization, and overall terminal performance [17, 15, 18].

Various simulation paradigms have been applied in terminal research. Discrete Event Simulation (DES) is the most frequently used approach because container terminal operations can naturally be represented as sequences of discrete operational events such as truck arrivals, gate processing, container handling, and equipment usage. Other approaches such as System Dynamics, Agent-Based Modeling, and microscopic traffic simulation have also been applied to investigate specific operational and strategic questions [19, 20, 18]. A more detailed discussion of simulation methodologies is provided in Section 2.3. Despite their widespread adoption, traditional simulation projects suffer from several limitations. Developing a simulation model typically requires extensive domain knowledge, detailed process understanding, and considerable modeling effort. The collection, formalization, implementation, calibration, and validation of process knowledge often represent the most time-consuming phases of a simulation project. Furthermore, manually developed simulation models most of the time rely on expert assumptions regarding process behavior. However, experts often describe how processes are intended to operate rather than how they are actually executed in practice, therefore, unintended discrepancies can appear between the simulation model and the real operational processes [5].

In dynamic environments like container terminals, these differences can impact the quality of simulation results. Another challenge is the continuous maintenance and calibration effort required to keep simulation models aligned with changing operational conditions. As terminal processes, workloads, and operating rules evolve over time, manually maintaining simulation models becomes costly and requires technical suppoert. At the same time, modern container terminals continuously record large amounts of operational event data through Terminal Operating Systems, Gate Operating Systems, and related information systems. These data sources provide detailed information about actual process executions and operational behavior. Rather than relying mainly on expert knowledge, process knowledge could be extracted directly from operational data. Process Mining provides a methodological framework for discovering, analysing, and monitoring real processes based on historic operational data. By deriving process information directly from recorded datasets, Process Mining offers the potential to reduce modeling efforts, while improving the representation of real-world behaviour. Consequently, it represents a promising foundation for data-driven simulation model construction and automated simulation model discovery.

2.2 Operational Data and Event Logs

2.2.1 Transactional Information Systems

Woraus und warum entstehen überall überhaupt Everntlogs? Today, most business processes are supported and documented through information systems. These systems are primarily designed to facilitate and document operational activities rather than to analyze whole process flows. Examples include Enterprise Resource Planning (ERP) systems, Customer Relationship Management (CRM) systems, Manufacturing Execution Systems (MES), and Terminal Operating Systems (TOS) used in container terminals. Whenever users or objects interact with such systems, digital records are generated and stored. Activities such as creating, updating, processing, or completing an order leave traces in the underlying databases. While this information is mainly required for operational control, documentation, and basic analysis purposes, it simultaneously creates detailed records of (partial) process executions. In container terminals, operational data are continuously generated by systems such as the Terminal Operating System (TOS), Gate Operating System (GOS), and Yard Management System (YMS). Typical examples include the registration of a truck arrival at a gate, the assignment of a container to a storage location, or the execution of a container handling operation. Each transaction recorded by these systems creates information about what happened, when it happened, and which operational object was involved. As those kind of business processes are executed step by step, these recorded transactions form chronological sequences of events. Such event data provide the foundation for Process Mining techniques, which aim to reconstruct and analyze actual process behavior directly from system-generated data. Regarding the HHLA-CTB information systems it should be noted, that due to historical growth and development of the individual systems, not all systems collect the same level of detail or have alternating names for the same facilities, which will be elaborated on when discussing the event log engineeing.

2.2.2 Events, Cases and Traces

Explanaiton of Event Case Trace Activity Timestamp Attribute Using HHLA Examples

Process Mining views business processes as sequences of events recorded during one process execution. To describe these sequences, several fundamental concepts should be explained.

Event An event represents the occurrence of a specific activity within a process at a particular point in time. Each event documents that something happened during the execution of a process. Examples of events in a container terminal include:

- Truck drives through OCR gate,
- Gate Check completed,
- Truck arrives at yard position,
- Container operation at yard position completed,
- Truck exits the terminal.

Activity An activity describes the type of action that is being executed. Multiple events can belong to the same activity while occurring at different times or for different process instances. Examples of activities are:

- Gate In,
- Container-Delivery,
- Container-Receive,
- Gate Out.

Case A case represents a single process instance. It is defined by the object whose process execution is being analyzed. Depending on the application domain, a case can represent different entities, such as:

- a customer order,
- a container,
- a truck visit.

In the context of this thesis, a truck visit serves as the primary process instance and therefore represents the case thats being investigated in.

In traditional event logs, events are often represented by a single timestamp, typically indicating the completion of an activity. However, for performance-oriented Process Mining and simulation applications, additional lifecycle information is required. A process activity can generally be associated with three relevant temporal states:

- **Enabled Timestamp**: The moment an activity becomes executable and starts waiting for the required resource.
- **Start Timestamp**: The moment execution of the activity actually begins.
- **Complete Timestamp**: The moment execution of the activity finishes.

The distinction between these timestamps enables the separation of waiting time and service time. Waiting time corresponds to the duration between the enabled and start timestamps, whereas service time corresponds to the duration between the start and completion timestamps. Such lifecycle information is particularly important for Process Mining-based simulation approaches, as it allows resource contention and queueing effects to be represented explicitly.

ATTRIBUTES In addition to the mandatory process information, events may contain supplementary attributes that provide further context about the execution of the process. Examples of event attributes in container terminal operations include:

- Truck ID,
- Container Number,
- Shipping Line,
- Yard Block,
- Container Type,
- Equipment Identifier,
- Trucking Company.

These attributes enable more detailed analyses and are particularly important for data-aware Process Mining and simulation approaches.

Trace A trace is the chronologically ordered sequence of all events belonging to a specific case. For a truck visit, a trace may appear as follows:

> OCR Gate Entry → Gate In → Yard Arrival → Container Delivery → Container Pick-Up → Gate Out

A trace therefore describes the actual execution path followed by a single process instance.

2.2.3 Event Logs

An event log is the central data structure used in Process Mining. It consists of a collection of events that are grouped into cases and ordered according to their timestamps. At a minimum, each event in an event log contains three essential attributes:

- Case Identifier,
- Activity,
- Enabled Timestamp,
- Start Timestamp
- Complete Timestamp
- Resource
- Truck Type

Additional attributes are usually included to enrich the event data with operational context. Table 2.1 illustrates a simplified event log.

Table 2.1: Simplified Event Log Example

Case ID	Activity	Timestamp	Yard Block
T001	Gate In	09:13:04	-
T001	Yard Arrival	09:21:46	B04
T001	Gate Out	09:42:18	-
T002	Gate In	09:17:12	-
T002	Yard Arrival	09:30:51	B08
T002	Gate Out	09:58:33	-

For data exchange between Process Mining tools, event logs are commonly stored in standardized formats. The most widely adopted standard is the eXtensible Event Stream

(XES) format. XES was specifically developed for Process Mining applications and enables the storage of events, cases, timestamps, and additional attributes in a structured and tool-independent way. Event logs provide the link between operational information systems and Process Mining techniques. By organizing recorded events into cases and traces, they transform raw operational data into a representation that makes process discovery, conformance checking, and performance analysis possible.

2.3 Process Mining Foundations

2.3.1 Definitions

Organizations increasingly collect large volumes of operational data through enterprise systems such as Enterprise Resource Planning (ERP), Customer Relationship Management (CRM), and Workflow Management Systems (WfMS). Various analytical disciplines have emerged to exploit these data sources. Among the most relevant are Business Intelligence, Data Mining, and Process Mining. Business Intelligence (BI) focuses primarily on descriptive analytics and the monitoring of organizational performance through dashboards, reports, and key performance indicators (KPIs). The goal of BI is to provide decision makers with aggregated information about past and current business performance [21]. Data Mining aims to identify patterns, correlations, classifications, or predictive relationships within large datasets. Traditional data-mining approaches treat observations largely as independent records and often disregard the ordering and dependencies between events [21]. Process Mining bridges the gap between traditional model-based process analysis and data-driven analytics. It exploits event data recorded by information systems to discover, monitor, and improve business processes. Instead of focusing on isolated records, Process Mining explicitly considers the sequence of events that constitute process executions. Van der Aalst defines Process Mining as a discipline that aims to improve operational processes through the systematic use of event data and process models [21]. A central characteristic of Process Mining is that it combines process models with observed execution data. This enables analysts to discover actual process behaviour, compare observed executions with expected behaviour, identify bottlenecks, and derive insights for process improvement [21].

Process Perspectives

Business processes can be analysed from different perspectives. Early Process Mining research distinguished four primary perspectives that together provide a comprehensive view of process execution [5].

Control-Flow Perspective The control-flow perspective describes the ordering relations between activities and captures how process instances progress through a process. It focuses on questions such as which activities occur, in which order, and which alternative execution paths exist.

Time Perspective The time perspective focuses on temporal process characteristics such as activity durations, waiting times, throughput times, and bottlenecks. The analysis of timestamps contained in event logs enables the identification of performance issues and delays.

Resource Perspective The resource perspective investigates the organizational dimension of a process. It considers which resources execute activities, how work is distributed among employees, and how organizational structures influence process behaviour.

Data Perspective The data perspective analyzes process-related attributes and case characteristics. Examples include customer types, product categories, order volumes, or risk classes. Such attributes often influence process decisions and execution paths [5]. Together, these perspectives provide a multidimensional view of business processes and form the foundation of advanced process analysis techniques.

Process Discovery

Process discovery is the most prominent branch of Process Mining. Its objective is to automatically construct a process model solely from recorded event data without prior process knowledge [21]. The input to process discovery is an event log containing historical process executions. The output is a process model that aims to reproduce the observed behaviour. Depending on the chosen discovery algorithm, the resulting model may be represented as a Petri net, process tree, BPMN model, or directly-follows graph. Over the years, numerous discovery algorithms have been proposed. Three classical examples are:

- **Alpha Miner**: One of the earliest process discovery algorithms, capable of identifying causal dependencies and parallel behaviour from event logs.
- **Heuristic Miner**: An extension of Alpha Miner that incorporates frequency information and is more robust towards noise and infrequent behaviour.
- **Inductive Miner**: A modern discovery algorithm that recursively decomposes event logs and guarantees the production of sound process models. It is widely used due to its robustness and ability to balance precision and simplicity.

Process discovery enables organizations to obtain an objective representation of how their processes are actually executed rather than how they are presumed to operate [21].

Conformance Checking

While process discovery focuses on constructing process models, conformance checking focuses on comparing observed process executions with an existing process model [22]. The objective is to identify deviations between the model and reality. Such deviations may indicate compliance violations, process inefficiencies, exceptional cases, or outdated process documentation. The quality of discovered and analyzed process models is commonly evaluated using four criteria [22, 21]:

- **Fitness**: Measures the extent to which observed process executions can be replayed by the process model.
- **Precision**: Evaluates whether the model avoids allowing excessive behaviour that was not observed in reality.
- **Generalization**: Assesses whether the model captures underlying process behaviour beyond the specific instances present in the event log.
- **Simplicity**: Favours models that explain observed behaviour with minimal complexity while remaining understandable.

These criteria provide a balanced framework for evaluating process model quality and form the basis of many process-mining evaluation approaches.

Process Enhancement

The Process Mining literature commonly distinguishes three major categories of process mining techniques: process discovery, conformance checking, and process enhancement [21]. Whereas discovery creates a process model from event logs and conformance checking compares observed behaviour with existing models, process enhancement aims to improve existing process models using information extracted from event data. Enhancement may take different forms. For example, performance information such as waiting times or activity durations can be projected onto a process model. Likewise, resource assignments, bottlenecks, and workload information can be incorporated to create richer process representations. Process enhancement therefore enables organizations not only to understand how processes operate, but also to identify opportunities for process improvement and optimization [21].

2.4 Business Process Simulation

2.4.1 Fundamentals of Business Process Simulation

Business Process Simulation (BPS) is a widely established analytical technique used to reproduce and analyze the dynamic behavior of business processes within a virtual environment. Rather than experimenting directly on operational systems, organizations build executable models that represent essential process characteristics and repeatedly execute these models under controlled conditions. This allows process behaviour to be observed without interfering with live operations [23]. Van der Aalst defines simulation as a flexible approach for analysing business processes and evaluating alternative operational scenarios. By recreating process executions under varying conditions, simulation can estimate process performance indicators such as throughput times, waiting times, resource utilization, service levels, and operational costs before changes are implemented in practice [23]. The primary purpose of simulation is decision support. Business processes are often characterized by uncertainty, dynamic resource interactions, and complex dependencies that make analytical evaluation difficult. Simulation provides a safe and cost-efficient environment for assessing operational changes while avoiding the risks associated with real-world experimentation [23]. One of the most important applications of simulation is *what-if analysis*. What-if analyses evaluate how process performance changes under alternative assumptions or operational configurations. Typical examples include increasing the number of available resources, modifying routing rules, changing workload levels, or introducing new operational policies. By comparing simulation outcomes across different scenarios, decision makers can assess the expected consequences of proposed changes before implementation [23]. Because of its ability to represent dynamic process behaviour and operational dependencies, simulation has become an established decision-support tool in fields such as manufacturing, healthcare, logistics, supply-chain management, and business process management. More recently, Process Mining research has increasingly integrated simulation with event data, enabling simulation models to be discovered and calibrated directly from recorded process executions [5, 13].

2.4.2 Business Process Simulation Models

A simulation model is a simplified representation of a real-world process. Although specific modeling approaches vary, most business process simulation models consist of several fundamental perspectives that jointly determine process behaviour during simulation runs [23, 5].

Control Flow

The control-flow perspective describes the behavioural structure of the process. It determines which activities exist, in which order they may occur, and which alternative execution paths are possible. In simulation models, the control-flow model serves as the structural backbone that governs process execution. Common representations include Petri nets, BPMN models, process trees, and workflow graphs [21, 23].

Arrival Process

The arrival process defines how new process instances enter the system. In business processes, this typically corresponds to customer requests, orders, applications, service tickets, or other incoming work items. Arrival behaviour is commonly represented through statistical distributions that describe the time between consecutive process arrivals. Since workload levels directly influence waiting times and resource utilization, realistic arrival modelling is essential for meaningful simulation results [23, 5].

Resources

Resources represent the entities that execute process activities. Depending on the application domain, resources may correspond to employees, machines, vehicles, cranes, terminal equipment, or information systems. Resource models typically include information about availability, capacities, execution capabilities, working schedules, and allocation policies. Resource constraints are often a major determinant of process performance because multiple process instances compete for limited resource capacity [23].

Activity Durations

The timing perspective describes how long activities require for execution. Typical simulation models distinguish between processing times, waiting times, and overall case durations. Activity durations are frequently represented through probability distributions derived from historical observations. These temporal characteristics largely determine process performance and throughput behaviour [23, 5].

Decision Logic

Many business processes contain decision points where alternative execution paths may be selected. Decision logic determines how these routing decisions are made during simulation. Traditional simulation models commonly employ static branching probabilities that specify how often a particular path is chosen. More advanced approaches model decisions as functions of contextual information, process attributes, or operational conditions [5, 7]. Together, control flow, arrivals, resources, timing information, and decision logic define the behaviour of a simulation model. These perspectives form the foundation of both traditional manually developed simulation models and modern simulation-discovery approaches discussed later in this chapter.

2.4.3 Simulation-Based Decision Support

Container terminals operate in highly dynamic environments characterized by fluctuating demand, limited resource capacity, and strong interdependencies between operational processes. Managers must continuously make decisions regarding equipment allocation, workforce planning, storage strategies, truck appointment policies, and capacity utilization while maintaining uninterrupted terminal operations. Direct experimentation in a live terminal environment is often impractical because operational changes may negatively affect service quality, throughput, or customer satisfaction. As a consequence, simulation has become one of the most widely applied decision-support approaches in container terminal research [24, 17, 18]. By providing a virtual representation of terminal operations, simulation enables decision makers to evaluate operational alternatives before implementation. Typical applications include:

- evaluation of truck appointment policies,
- resource and workforce planning,
- capacity planning and investment decisions,
- yard layout and storage strategies,
- yard crane allocation policies,
- congestion analysis and bottleneck identification,
- equipment scheduling and maintenance planning.

A central advantage of simulation is its ability to support what-if analyses. Alternative operational scenarios can be evaluated under identical conditions, allowing direct comparison of performance measures such as truck turnaround times, resource utilization, throughput, waiting times, and service levels. This capability is particularly valuable in container terminals because small operational changes frequently generate system-wide effects due to the strong interdependencies between gate operations, yard processes, and vessel handling activities [3, 24, 17]. For truck operations, simulation has been extensively applied to study the effects of Truck Appointment Systems (TAS), gate capacities, yard utilization, and equipment allocation strategies. Such analyses support terminal operators in balancing efficiency improvements with operational robustness under uncertain workloads and changing operating conditions [15, 16].

2.4.4 Simulation Model Verification, Calibration and Validation

The usefulness of a simulation model depends not only on its structure but also on its ability to represent real-world behaviour with sufficient accuracy. Consequently, simulation projects typically include activities related to verification, calibration, and validation [23].

Verification Verification addresses the question:

> "Was the simulation model implemented correctly?"

The objective of verification is to ensure that the executable simulation model correctly reflects the conceptual model and contains no implementation errors. Verification therefore focuses on the internal consistency of the model rather than its correspondence with reality [23].

Calibration Calibration concerns the estimation and adjustment of simulation parameters so that the model accurately reflects observed operational conditions. Typical calibration tasks include fitting arrival distributions, activity durations, resource capacities, and routing probabilities using historical observations. Calibration therefore links the conceptual model with empirical process data [5, 13].

Validation Validation addresses the question:

> "Does the simulation model adequately represent the real system?"

Validation evaluates whether simulation outputs correspond sufficiently well with observed process behavior. This often involves comparing simulated and real performance measures such as throughput times, waiting times, resource utilization, or process paths. A validated model provides confidence that simulation results can be used for decision support and process analysis [23, 5]. Historically, these activities have been performed manually as part of traditional simulation studies. However, the emergence of simulation discovery and Process Mining has increasingly shifted verification, calibration, and validation towards data-driven approaches. Automatically discovered simulation models can use historical event data both for parameter estimation and for systematic comparisons between simulated and observed process behavior [5, 13, 25].

2.5 Simulation Discovery and Data-Aware Simulation

Here you should recognise the components from "Business Process Simulation"

The development of simulation models traditionally relies on a lot of manual modeling efforts and substantial domain expertise. Process analysts and simulation experts typically construct process models, resource configurations, routing probabilities, and performance parameters based on interviews, observations, and historical reports. [SOURCE] This approach is intuitive and expertise based, but often costly, time-intensive, and prone to subjective assumptions. To reduce modeling effort and alignment with observed process behavior, Process Mining research focuses on the automatic discovery of simulation models directly from historic process data in the form of event log. This research stream is commonly referred to as *simulation discovery*.

2.5.1 From Manual Simulation to Simulation Discovery

Traditional simulation projects require process experts to manually define control-flow structures, estimated activity durations, planned resource assignments, and routing probabilities. Consequently, the resulting simulation models often reflect how processes are expected to operate rather than how they are actually executed in practice [5]. Furthermore, simulation models require continuous maintenance and recalibration to remain aligned with changing operational conditions. One of the first comprehensive approaches to automatically derive simulation models from event logs was proposed by Rozinat et al. [5]. Their framework combines Process Mining techniques with simulation modelling and automatically discovers multiple simulation perspectives, including process control-flow, activity durations, resource assignments, and branching probabilities. By deriving simulation parameters directly from historical event data, the resulting models can more accurately reflect real process executions while reducing modelling effort. Building upon this idea, subsequent research increasingly automated simulation model construction. Camargo et al. [13] introduced a framework for the automated discovery of business process simulation models from event logs. Their approach systematically extracts simulation parameters and integrates them into executable simulation models. More recent work further extends simulation discovery through machine learning and deep learning techniques, enabling simulation models to capture complex behavioural patterns directly from operational data. While these approaches often improve predictive accuracy, they may also reduce transparency and configurability compared to traditional simulation models. As a result, simulation discovery has emerged as an important research area at the intersection of Process Mining and Business Process

Simulation, aiming to automatically generate simulation models that faithfully represent observed process behaviour while minimizing manual modelling effort.

2.5.2 Discovery of Simulation Parameters

Simulation models consist of multiple perspectives that must be discovered and calibrated from historical event data. Modern simulation discovery approaches therefore decompose the discovery problem into several parameter categories.

Control-Flow Discovery

Control-flow discovery aims to reconstruct the behavioural structure of a process from event logs. Process discovery algorithms such as the Inductive Miner derive formal process models, commonly represented as Petri nets, BPMN models, or process trees. The discovered process model serves as the behavioural backbone of the simulation model and defines the possible execution paths that process instances can follow [21, 5].

Arrival Discovery

Arrival discovery focuses on modelling how new process instances enter the system. In simulation models, this typically requires estimating the inter-arrival times between subsequently arriving cases. Arrival patterns are commonly derived from historical timestamps and represented through statistical distributions or predictive models. Accurate arrival modelling is essential because workload dynamics strongly influence waiting times, queue lengths, and resource utilization [5, 10].

Resource Discovery

Resource discovery concerns the identification of the resources involved in process execution and the assignment of activities to these resources. Resource-aware simulation models additionally require information about organizational structures, resource calendars, availability patterns, and allocation policies. Modern simulation-discovery approaches increasingly model resource behaviour as a context-dependent phenomenon instead of assuming static activity-resource mappings [26, 10].

Timing Discovery

Timing discovery aims to capture the temporal behaviour of process executions. Typical parameters include activity execution times, waiting times, service times, and overall throughput times. Early approaches estimate these parameters using fitted statistical distributions, whereas more recent approaches employ machine-learning techniques to model context-dependent temporal behaviour. Timing information is crucial for reproducing realistic process performance and resource interactions during simulation [5, 13, 10].

Decision Logic Discovery

Many business processes contain decision points at which alternative execution paths may be chosen. Traditionally, these decisions are modelled using branching probabilities derived from historical frequencies. Such probabilities reflect how often a specific path was observed in the event log. However, routing decisions in real processes are often influenced by process attributes, execution history, resource conditions, and operational context. Consequently, recent simulation-discovery approaches increasingly focus on discovering data-dependent decision rules that dynamically adapt simulation behaviour to the current process context [7, 8, 10].

2.5.3 Data-Driven and Data-Aware Simulation

Early simulation-discovery approaches typically represent decision-making behaviour using static branching probabilities. At a decision point, the probability of selecting a particular execution path is estimated from the observed frequency of that path in the event log. While such probabilities provide a simple representation of process behaviour, they cannot explain why different decisions occur under different circumstances [5]. To overcome these limitations, recent research has introduced data-aware simulation approaches. Instead of relying solely on global routing frequencies, process decisions are conditioned on contextual information describing the process instance. Mannhardt et al. [7] propose the concept of data-aware stochastic process models, where transition probabilities depend on process attributes rather than being represented as fixed probabilities. This allows simulation models to differentiate between cases with different characteristics and reproduce behaviour that depends on process data. Building upon this idea, de Leoni, Vinci, Leemans, and Mannhardt [8] introduce the concept of *process states*. A process state represents a data-aware abstraction of a process instance and may include process attributes, activity execution history, and other contextual information accumulated throughout process execution. Instead of assigning a single probability to an enabled transition, activity-firing probabilities become dependent on the current process state. The authors demonstrate that simulation models based on process-state-aware decision rules achieve higher behavioural accuracy than traditional frequency-based branching probabilities. This transition can be summarized as follows:

$$\begin{array}{c}\text{Static Branching Probabilities}\\ \downarrow \\ \text{Data-Aware Decisions}\\ \downarrow \\ \text{Process-State-Aware Simulation}\end{array}$$

The most recent developments extend this idea beyond control-flow decisions. In the Prosit framework, Vinci et al. [10] propose Decision-Aware Business Process Simulation models that discover control-flow decisions, execution times, waiting times, and resource behaviour using probabilistic decision trees. These models condition simulation parameters on contextual features such as trace attributes, process history, resource assignments, and workload-related information. An important distinction must be made between trace-level attributes and dynamically changing process-state information. The current Prosit framework primarily utilizes trace attributes, trace history, resource information, and workload-related features for simulation parameter discovery [10]. Arbitrary event-level contextual attributes that change continuously during process execution are not explicitly included in the published feature definitions. Nevertheless, the process-state perspective introduced by recent research suggests that richer operational context information may further improve simulation realism, particularly in complex operational environments such as container terminals. The evolution from frequency-based branching probabilities towards context-dependent and process-state-aware simulation models represents one of the most significant developments in simulation discovery research and provides the foundation for modern explainable and white-box simulation approaches [8, 10].

2.6 Explainable and White-Box Simulation

This chapter is suppose to explain why explainability is becoming a major research problem in simulation discovery, and then positioning Prosit as the current state-of-the-art answer *to that problem.*

2.6.1 Explainability in Simulation-Based Decision Support

Business Process Simulation is increasingly used to support operational and strategic decision making. Simulation results often form the basis for resource-allocation decisions, process redesign initiatives, capacity planning, and investment evaluations. Consequently, decision makers must be able to understand why a simulation model produces a particular outcome and how simulation behaviour changes when process parameters are modified. Traditional Business Process Simulation models are inherently transparent. Their control-flow structures, routing probabilities, resource allocations, and timing assumptions are manually specified and can therefore be inspected and modified by domain experts. As a result, simulation users can directly understand the assumptions underlying the generated simulation results [23]. Recent advances in simulation discovery increasingly employ machine-learning and deep-learning techniques to improve simulation accuracy. These approaches learn behavioural patterns directly from historical event data and often outperform manually calibrated models in terms of predictive capabilities [13, 27]. However, the resulting models frequently behave as black-box systems. While they can generate accurate predictions, the internal decision mechanisms that govern simulation behaviour are not readily interpretable by human analysts. This creates a tension between predictive accuracy and explainability. Black-box simulation models may successfully reproduce historical process behaviour while simultaneously limiting the analyst's ability to understand why particular routing decisions, waiting times, or resource allocations are generated. Consequently, the simulation model becomes difficult to inspect, configure, validate, and trust for operational decision support [11]. The explainability problem becomes particularly relevant in what-if analyses. Business Process Simulation is not only intended to reproduce observed behaviour but also to evaluate hypothetical future scenarios. Such analyses require analysts to intentionally modify process parameters, resource settings, or decision rules. In black-box simulation approaches, the underlying behavioural logic is often inaccessible, making it difficult to assess whether the simulated reactions to process changes remain realistic and trustworthy [11]. To address these limitations, recent research has started investigating explainable simulation models that combine the predictive capabilities of machine-learning approaches with transparent and interpretable decision logic. The goal is not only to reproduce process behaviour accurately but also to expose the reasoning mechanisms that determine simulation outcomes. Explainability therefore becomes an important prerequisite for trustworthiness, configurability, and practical adoption of data-driven simulation models.

2.6.2 Probabilistic White-Box Simulation Models

The concept of white-box simulation emerged as a response to the increasing reliance on opaque machine-learning models for simulation parameter discovery. White-box simulation models aim to make behavioural decisions explicitly visible and interpretable while maintaining a high degree of simulation accuracy. A key motivation behind white-box simulation is that business analysts must be able to inspect, understand, and modify the simulation behaviour that drives decision-support outcomes. Unlike black-box predictors, white-box approaches expose the discovered decision rules and allow analysts to evaluate the causal relationships that influence simulation parameters. An important step towards explainable simulation is represented by the Runtime Integration of Machine Learning and Simulation (RIMS) framework proposed by Meneghello et al. [11]. RIMS combines traditional discrete-event simulation with predictive machine-learning models at runtime. The authors explicitly argue that purely deep-learning-based simulation approaches suffer from limited explainability and restricted support for what-if analyses. By maintaining an explicit simulation model while integrating learned predictions during execution, RIMS seeks to combine simulation transparency with data-driven prediction capabilities [11]. More recently, Vinci et al. introduced the ProSiT framework (*PROcess SImulation Tool*), which represents one of the first dedicated white-box simulation-discovery approaches [10]. ProSiT extends data-aware simulation discovery by replacing opaque prediction mechanisms with probabilistic decision trees. These trees are used to represent multiple simulation dimensions including:

- control-flow decisions,
- activity execution times,
- waiting times,
- resource assignment behaviour,
- arrival processes.

The central idea of ProSiT is that simulation behaviour should remain directly interpretable. Instead of producing predictions through hidden neural-network representations, behavioural rules are exposed explicitly in the form of decision-tree structures. Analysts can therefore inspect, understand, and modify the conditions influencing simulation behaviour [10]. An additional contribution of ProSiT is the integration of uncertainty through probabilistic decision trees. Traditional decision trees often generate deterministic predictions. In contrast, probabilistic decision trees capture distributions of possible outcomes and thereby preserve the stochastic nature required for realistic process simulation [10]. This is particularly important because simulation models aim not only to reproduce average behaviour but also the variability observed in real operational environments. Experimental evaluations presented by Vinci et al. indicate that probabilistic white-box models can achieve simulation accuracy comparable to, and in some cases exceeding, alternative black-box approaches while simultaneously improving interpretability and configurability [10]. Furthermore, the framework enables process analysts to directly edit discovered rules and evaluate modified decision scenarios without retraining complex machine-learning models. Despite these promising results, research on white-box simulation remains at an early stage. Current evidence is largely limited to benchmark datasets and business-process environments investigated by the authors themselves. Large-scale empirical evaluations across different domains remain scarce. In particular, little evidence currently exists regarding the behaviour of white-box simulation models in highly dynamic cyber-physical systems such as container terminals. Moreover, explainability itself remains difficult to quantify. While decision trees are generally considered more transparent than neural networks, there is currently no universally accepted framework for measuring the practical interpretability of simulation-discovery models. Existing evaluations primarily focus on simulation accuracy, leaving questions related to user trust, decision-support quality, and operational usefulness largely unexplored. Consequently, white-box simulation should not yet be regarded as a mature solution to the explainability challenge. Rather, it represents an emerging research direction that attempts to balance three competing objectives:

(a) simulation accuracy,

(b) simulation explainability,

(c) simulation configurability.

The ProSiT framework currently represents one of the most comprehensive realizations of this vision and serves as the methodological foundation for the simulation approach investigated in this thesis [10].

2.7 Simulation Quality, Validation/Assessment, and Trustworthiness

This part is suppose to answer the question: How do we know whether these automatically discovered simulation models are actually good enough to support decisions?

As simulation models increasingly become automatically generated from event data, questions regarding their quality, reliability, and practical usefulness become increasingly important. While traditional simulation projects rely heavily on expert validation and manual calibration, simulation-discovery approaches automatically derive model structures and parameters directly from event logs. Consequently, the assessment of simulation quality has evolved into an active research area at the intersection of Process Mining and Business Process Simulation. Ultimately, the objective of a simulation model is not merely to reproduce historical data, but to support operational decision making. Therefore, simulation quality must be evaluated from multiple perspectives, including behavioural realism, temporal accuracy, resource representation, and decision-support capability.

2.7.1 Dimensions of Simulation Quality

Simulation models are multi-dimensional representations of real-world processes. Consequently, simulation quality cannot be reduced to a single performance indicator. Instead, different aspects of process behaviour must be evaluated separately.

Control-Flow Quality The control-flow perspective evaluates whether the simulation model reproduces the behavioural structure observed in reality. A high-quality model should generate process variants and activity sequences that resemble the execution behaviour recorded in the event log. Deviations may indicate missing process paths, unrealistic routing behaviour, or overgeneralized models [5, 28].

Temporal Quality Temporal quality measures how accurately the simulation reproduces timing-related process characteristics such as activity durations, waiting times, throughput times, and overall case durations. Since decision-support applications frequently focus on performance improvement, temporal realism is often considered a critical component of simulation quality [23, 28].

Resource Quality Resource quality evaluates how accurately simulation models represent resource behaviour, including resource assignments, workloads, utilization patterns, and capacity constraints. Resource-related inaccuracies can significantly affect simulation outcomes because waiting times and bottlenecks are often produced by resource contention rather than control-flow decisions alone [23, 10].

Decision-Support Capability A technically accurate simulation model does not automatically provide useful decision support. Simulation results must remain sufficiently realistic under changed conditions and alternative scenarios. Consequently, simulation quality also includes the ability of a model to support meaningful what-if analyses and generate trustworthy recommendations for operational decision making [28, 11].

2.7.2 Simulation Validation

Traditional simulation validation typically compares selected performance indicators between observed and simulated systems. However, modern simulation-discovery approaches require more comprehensive evaluation techniques because behavioural realism, timing characteristics, and workload dynamics must all be assessed simultaneously. Recent Process Mining research therefore introduced a variety of similarity metrics designed to compare event logs generated by simulation models with real-world event logs [28].

Control-Flow Similarity Metrics

Control-flow similarity focuses on comparing the behavioural structure of simulated and observed process executions. One of the most widely used techniques is the Earth Mover's Distance (EMD), originally introduced for stochastic conformance checking by Leemans et al. [29]. EMD measures how much probability mass must be moved between two trace distributions to transform one distribution into another. A lower EMD indicates stronger behavioural similarity. Building upon this idea, simulation-specific evaluation frameworks frequently employ Control-Flow Log Distance (CFLD) metrics. CFLD compares the frequencies of process variants generated by a simulation model against the variants observed in reality. This enables the assessment of behavioural realism at the process level [28]. Another common family of control-flow measures is based on N-Gram analysis. Rather than comparing complete traces, N-Gram methods compare local activity patterns consisting of consecutive activity sequences of length n. This allows behavioural similarities to be identified even when complete process variants differ [28].

Temporal Similarity Metrics

Control-flow similarity alone is insufficient because simulation models are frequently used to predict temporal process performance. Case Throughput Distance (CTD) compares throughput time distributions between simulated and real event logs. The metric evaluates how accurately a simulation reproduces the overall duration of process instances [28]. Absolute Error Distance (AED) and Relative Error Distance (RED) quantify the discrepancy between observed and simulated temporal characteristics. These metrics are commonly used to assess waiting times, service times, and throughput times at different levels of aggregation [28]. Temporal metrics are particularly relevant in operational environments where process performance directly affects service quality. In container terminal operations, for example, accurate prediction of truck turnaround times and waiting times represents a central requirement for practical simulation-based decision support.

Arrival and Congestion Metrics

In many operational systems, process performance is strongly influenced by workload levels and arrival dynamics. Consequently, simulation accuracy must also be evaluated with respect to congestion behaviour and workload generation. The Case Arrival Rate (CAR) metric compares the arrival patterns of process instances between simulated and observed event logs. Accurate arrival modelling is essential because unrealistic workload generation can distort resource utilization, queue lengths, and waiting times [28]. For container terminal operations, arrival behaviour is particularly important because truck arrivals and yard workloads directly influence congestion levels throughout the terminal. A simulation model that accurately reproduces individual process traces but fails to replicate realistic truck arrival dynamics may still generate misleading decision-support results.

Wasserstein Distance

Many similarity metrics used in simulation validation are based on the Wasserstein Distance, which originated in optimal transport theory. The Wasserstein Distance measures the minimum effort required to transform one probability distribution into another. Intuitively, it can be interpreted as the amount of work required to relocate probability mass between distributions. Earth Mover's Distance represents a discrete formulation of the Wasserstein Distance and is particularly well suited for comparing trace distributions and temporal distributions in Process Mining applications [29]. Due to its ability to compare complete distributions rather than only summary statistics, Wasserstein-based metrics have become a central component of modern simulation validation frameworks.

2.7.3 Trustworthiness of Automatically Discovered Simulation Models

As simulation models become increasingly data-driven and automatically generated, simulation quality alone is no longer sufficient. Decision makers must also be able to trust the simulation model and understand its limitations. Recent research therefore increasingly focuses on the trustworthiness of simulation-discovery approaches. Trustworthiness encompasses multiple dimensions, including behavioural accuracy, robustness, explainability, reproducibility, and decision-support reliability [11, 10]. The framework proposed by Chapela-Campa et al. represents one of the first systematic efforts to measure simulation quality across multiple dimensions simultaneously [28]. Their work highlights that simulation accuracy depends on different behavioural perspectives and that no single validation metric is sufficient to judge model quality comprehensively. Similarly, recent healthcare studies using the ProSiT framework demonstrate that improved behavioural realism can positively influence simulation-based resource-allocation decisions [12]. However, these studies remain largely confined to healthcare and administrative business processes. Although promising results have been reported, empirical evidence regarding the robustness and trustworthiness of automatically discovered simulation models remains limited. Most evaluations focus on reproducing historical behaviour rather than assessing simulation performance under substantially changed operational conditions. Consequently, important questions regarding model stability and decision-support reliability remain only partially addressed.

2.7.4 Research Gap

Container terminals represent highly dynamic and resource-constrained operational environments characterized by complex interactions between truck processes, yard operations, storage systems, and handling equipment. As a result, simulation has become one of the most widely used analytical methods for supporting operational decision making in container terminal management [24, 17, 18]. Despite this extensive use of simulation, most container terminal simulation models continue to be developed manually. Model construction, calibration, and maintenance require substantial modelling effort and domain expertise, limiting scalability and practical applicability in rapidly changing operational environments [5]. Simulation discovery addresses this limitation by automatically generating simulation models from event logs. Over the past decade, significant advances have been achieved in the discovery of simulation parameters, data-aware decision logic, and automated model construction [13, 25, 10]. More recently, data-aware simulation approaches demonstrated that contextual information and process-state representations can substantially improve simulation accuracy compared to traditional branching probabilities [7, 8]. In parallel, white-box simulation approaches such as ProSiT introduced interpretable probabilistic decision models that aim to improve transparency and configurability while maintaining strong predictive performance [10]. However, existing empirical evaluations have primarily been conducted in healthcare, administrative, and generic business-process environments. To date, there is limited evidence regarding the applicability and behaviour of simulation-discovery approaches within large-scale container terminal operations. Furthermore, the increasing complexity of data-aware and decision-aware simulation models raises unresolved questions concerning:

- the robustness of automatically discovered simulation models,
- their trustworthiness for operational decision support,
- their ability to reproduce congestion effects and workload dynamics,
- the practical value of white-box simulation in complex logistics environments.

This thesis does not aim to provide a general answer to these broader research challenges. Rather, it contributes an empirical case study within the context of truck-handling operations at the HHLA Container Terminal Burchardkai (CTB). By evaluating simulation-discovery approaches in a real-world container terminal environment, the thesis provides evidence regarding their applicability, strengths, and limitations in a domain that has received comparatively little attention within the existing simulation-discovery literature.

does this give of: I am not solving trustworthiness of simulation discovery here! I am providing a container-terminal case study that contributes evidence toward that larger research question.

3
Research Methodology (12)

Warum

Wie

Mit welchem Forschungsdesign

3.1 Research Design

Case Study Design Science / Experimental Evaluation Quantitative Evaluation

3.2 REsearch Process

Visual: Event Log → Discovery → Simulation → Experiments → Evaluation

3.3 Case Study Environment

HHLA

Terminal Truck Process IT Landscape

3.4 Data Collection

Source Systems Extraction

3.5 Event Log Construction

3.5.1 Raw Data Sources

yard interpolation truck demand interpolation

3.5.2 Case Definition

3.5.3 Activity Definition

Resource Pooling

3.5.4 Timestamp selection

Waiting time generation

3.5.5 Context Attributes

Context/ Container Feature Engineering

3.6 Evaluation Methodology

No results, just what metrics are being used (explained in the backgropund chapter), which comparison Logic and what scenarios are being built.

4

Simulation Discovery (12)

4.1 Process Discovery Results

4.1.1 Discovered COntrol Flow

(Colored?) Petri Net Process Model

4.1.2 Process CHaracteristics

Variants Loops

Frequences

4.2 Simulation Parameter Discovery

4.2.1 Arrival Processes

4.2.2 Resource MOdels

4.2.3 Temporal Models

4.2.4 Decision Models

4.3 Contextual Attribute Analysis

Have context features given a benefit to the model?

4.3.1 Attribute Importance

target Block outvoted etc. Visit Type

4.3.2 Process State Analysis

4.4 Resulting Simulation Model

What Attributes are importnant and what resources are discovered?

5

Experimental Setup (8)

Here we simulate

5.1 Baseline Configuration

5.2 Validation Scenarios

5.2.1 Historical Replay

5.2.2 Pertubation Scenarios

5.2.3 What-If Scenarios

5.3 Evaluation Metrics

NGD CTD CAR etc.

6

Results (18)

7 Discussion (10)

7.1 Implications for Container Terminals

7.2 Implications for Data-Aware Simulation

7.3 Trustworthiness of Simulation Discovery

Simulation Validity from Aalst 2015 in Business Process Simulation Survival Guide Trustworthiness Validierung Verifikation Grenzen von Simulation

7.4 Limitations

8 Conclusion (3)

This thesis addressed one of the most important open challenges in …

The theoretical contribution of this research was …

Finally, we demonstrated how our new framework…

Ultimately, this thesis provides both a…

8.1 Future Work

While the proposed framework establishes a solid foundation for… To advance this framework, future efforts should be directed across the following key areas:

- **Different TErminal Setting):** A fundamental structural difference between SG

8.2 Personal Takeaways

Beyond the theoretical conclusions of this thesis, I would like to close by reflecting on the personal and professional insights I take away from this research project:

(a) **Independent Project Management:** I learned how to self-organize, manage timelines effectively, and work independently to drive a large-scale project to completion.

(b) **Embracing Setbacks as Progress:** I understood firsthand that failure, unexpected results, and rigorous debugging are not roadblocks, but rather inherent and necessary steps in the scientific research process.

(c) **The Value of First Principles Thinking:** I learned that when facing complex challenges, the most effective strategy is often to step back, re-evaluate the fundamental mathematics and logic, and approach the problem from a renewed perspective.

(d) **Prioritizing Reproducibility Over Complexity:** I realized that a tested, reproducible, and well-documented framework holds far more scientific value than a system that is overly complex but unmanageable. Furthermore, working in small, modular, and heavily tested scripts proved absolutely necessary when scaling up to a large codebase.

(e) **Navigating Collaborative Research Dynamics:** I experienced the immense value of working within an active research group, where navigating open questions and integrating diverse feedback significantly enriched the final work.

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